

Aashish Gupta

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KEY PROJECTS

- Analysis of large-scale structures around planet-forming disks in DECO**
2024-present
I am leading the analysis of large-scale (≥ 500 au) gaseous structures connected to planet-forming disks, as observed in ALMA Large Program DECO (Disk-Exoplanet C/Onnection, PI: L. Ilseidore Cleeves).
- Development of TIPSy, a code to analyse streamer observations**
Second Ph.D. project at ESO, Germany with Dr. Anna Miotello; 2022-2023
I developed the first of its kind code to fit molecular-line observations of elongated trails of infalling gas, often called streamers, with theoretically expected trajectories of infalling gas.
- Using reflection nebulae to identify Class II sources undergoing late infall**
First Ph.D. project at ESO, Germany with Dr. Anna Miotello; 2021-2022
I demonstrated that association with reflection nebulae can be used to identify Class II sources accreting material from surrounding clouds, which further suggests that a significant fraction of Class II sources could be undergoing this phenomena.
- Studying interplay between molecular clouds and young stars in Rho Ophiuchus**
Master's thesis at National Central University, Taiwan with Prof. Dr. Chen, Wen-Ping; 2019-2021
I diagnosed the spatial and kinematic distribution of the young stellar population with respect to the deepest JCMT continuum emission map in the core of Rho Oph cloud complex.
- Investigating effects of magnetic fields on gas kinematics in protostars**
Academia Sinica Institute of Astronomy and Astrophysics (ASIAA), Taiwan internship with Dr. Yen, Hsi-Wei; 2020-2021
Using data from JCMT and SMA, I investigated correlations between magnetic field orientations/strengths and gas kinematics for a sample of ~ 50 protostars in Perseus molecular clouds.
- Pipeline for detection and analysis of stellar flares**
Internship at Aryabhata Research Institute of Observational Sciences (ARIES), India with Dr. Jeewan Pandey; 2018
I developed a pipeline to detect and analyze flaring signatures in Kepler's lightcurves.
- Fourier analysis of RR Lyrae variable stars**
ARIES summer internship with Dr. Sneha Lata; 2017
I developed a program to analyse time series data of RR Lyrae Variables and estimate their physical parameters.
- Development of sugar-based additive manufacturing machine**
Bachelor's research project at IIITDMJ, India with Prof. Puneet Tandon; 2016
I assisted in the design and fabrication of a machine for 3D printing objects out of pure sugar.

EDUCATION

Year	Qualification	Institute	Grade
2021-present	Doctor of Philosophy (Astronomy)	European Southern Observatory and Ludwig Maximilian University, Germany	1.2 (magna cum laude)
2019-2021	Master of Science (Astronomy)	National Central University, Taiwan	89.6/100
2015-2019	Bachelor of Technology (Mechanical Engineering)	PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur, India	8.1/10

SKILLS

- *Programming Languages:* Python, C/C++, Octave, HTML, Shell scripting
- *Softwares:* LaTeX (Typesetting), CASA (Radio data processing), IRAF (Optical data reduction), Git (Software management), DS9 (Imaging), CARTA (Radio imaging)
- *Languages:* Hindi (Native), English (Proficient, IELTS score 8), Mandarin Chinese (Basic speaking and listening), German (Basic, A1)

PUBLICATIONS

First author papers:

- *"TIPSY: Trajectory of Infalling Particles in Streamers around Young stars. Dynamical analysis of streamers around S CrA and HL Tau"*
Gupta A., Miotello A., Williams J. P., Birnstiel T., Kuffmeier M., Yen H.-W., 2024, A&A, 683, A133. doi:10.1051/0004-6361/202348007
- *"Reflections on nebulae around young stars. A systematic search for late-stage infall of material onto Class II disks"*
Gupta A., Miotello A., Manara C. F., Williams J. P., Facchini S., Beccari G., Birnstiel T., Ginski C., Hacar A., Kuffmeier M., Testi L., Tychoniec L., Yen H.-W., 2023, A&A, 670, L8. doi:10.1051/0004-6361/202245254
- *"Effects of Magnetic Field Orientations in Dense Cores on Gas Kinematics in Protostellar Envelopes"*
Gupta A., Yen H.-W., Koch P., Bastien P., Bourke T. L., Chung E. J., Hasegawa T., Hull C. L. H., Inutsuka S., Kwon J., Kwon W., Lai S.-P., Lee C. W., Lee C.-F., Pattle K., Qiu K., Tahani M., Tamura M., Ward-Thompson D., 2022, ApJ, 930, 67. doi:10.3847/1538-4357/ac63bc
- *"Interplay between Young Stars and Molecular Clouds in the Ophiuchus Star-forming Complex"*
Gupta A., Chen W.-P., 2022, AJ, 163, 233. doi:10.3847/1538-3881/ac5cc8

Co-author papers:

- *"Spatially correlated stellar accretion in the Lupus star-forming region: Evidence for ongoing infall from the interstellar medium"*
Winter A. J., Benisty M., ...**Gupta A.** et al., 2024, A&A, 691, A169, doi:10.1051/0004-6361/202452120
- *"Discovery of an Accretion Streamer and a Slow Wide-angle Outflow around FU Orionis"*
Hales A. S., **Gupta A.**, et al., 2024, ApJ, 966, 96, doi:10.3847/1538-4357/ad31a1
- *"Anatomy of the Class I protostar L1489 IRS with NOEMA: disk, streamers, outflow(s) and bubbles at 3mm"*
Tanious M., Le Gal R., ...**Gupta A.** et al., 2024, A&A, 687, A92, doi:10.1051/0004-6361/202348785
- *"A dusty streamer infalling onto the disk of a class I protostar. ALMA dual-band constraints on grain properties and mass infall rate."*
Cacciapuoti L., Macias E., **Gupta A.** et al., 2023, A&A, 682, A61, doi:10.1051/0004-6361/202347486
- *"The VLT MUSE NFM view of outflows and externally photoevaporating discs near the orion bar"*
Haworth T. J., Reiter M., ...**Gupta A.** et al., 2023, A&A, 670, L8. doi:10.1051/0004-6361/202245254
- *"The JCMT Transient Survey: Four-year Summary of Monitoring the Submillimeter Variability of Protostars"*
Lee Y.-H., Johnstone D., ...**Gupta A.** et al., 2021, ApJ, 920, 119. doi:10.3847/1538-4357/ac1679
- *"No Impact of Core-scale Magnetic Field, Turbulence, or Velocity Gradient on Sizes of Protostellar Disks in Orion A"*
Yen H.-W., Zhao B., Koch P. M., **Gupta A.**, 2021, ApJ, 916, 97. doi:10.3847/1538-4357/ac0723
- *"VR CCD Photometry of Variable Stars in the Globular Cluster NGC 4147"*
Lata S., Pandey A. K., ...**Gupta A.** et al., 2019, AJ, 158, 51. doi:10.3847/1538-3881/ab22a6

PARTICIPATION IN MEETINGS

Invited talks:

- StarPlan seminar, University of Copenhagen, Copenhagen, Denmark, December 2024
- Astronomy seminar, University of Exeter, Exeter, UK, February 2024
- Core2disk III workshop, Paris, France, October 2023
- Origins seminar at Chalmers University, Gothenburg, Sweden, August 2023
- ESO star and planet formation seminar, Garching, Germany, February 2023
- ECOGAL seminar, Online, February 2023
- MPE-CAS journal club on star and planet formation, Garching, Germany, July 2022

Contributed talks:

- Born in Fire: Eruptive Stars and Planet Formation, Santiago, Chile, September 2024
- New Heights in Planet Formation, Garching, Germany, July, 2024
- European Astronomical Society Annual Meeting (EAS), Krakow, Poland, July 2023
- Meeting of ALMA Young Astronomers (MAYA), Online, March 2023
- Annual Meeting of Astronomy Society of the Republic of China (ASROC), Taipei, Taiwan, September 2020

Posters:

- Inter+Stellar: Harnessing the Intersection Between Stars and the Interstellar Medium, Baltimore, USA, May 2025
- Star@Lyon conference, Lyon, France, June 2023
- Protostars and Planets VII, Kyoto, Japan, April 2023
- Annual Meeting of Astronomy Society of the Republic of China (ASROC), Taipei, Taiwan, September 2020
- Annual Meeting of the Physical Society of Taiwan (TPS), Pingtung, Taiwan, February 2020
- Exploring the Universe: Near Earth space science to extragalactic astronomy (EXPUNIV2018), Kolkata, India, Nov 2018

Workshops and schools:

- “DUSTBUSTERS school on protostellar discs and planet formation”, Italy, February 2022
- “7th KAGRA International Workshop”, NCU, Taiwan, December 2020
- “ALMA Imaging Workshop”, ASIAA, Taiwan, September 2020
- “ASIAA Summer Student Program”, ASIAA, Taiwan, July/August 2020
- “RAD@home Discovery Camp”, Nehru Planetarium, India, December 2018

AWARDS AND ACHIEVEMENTS

- VICO postdoctoral fellowship at University of Virginia, USA
- DFG funding for Ph.D. at ESO, Germany
- MOST scholarship for master’s program at NCU, Taiwan
- Award for academic performance at NCU, Taiwan
- Certificate of Merit for academic excellence at IIITDM Jabalpur, India
- Invitation to write an SMA Newsletter article on “Effects of Magnetic Field Orientations in Dense Cores on Gas Kinematics in Protostellar Envelopes”
- Poster presentation award in TPS 2020
- Winner of Astronomy Hackathon in Techkriti 2019; Winner of Cosima in Abhikalpan 2019; First runners up in Astronomy Quiz in Abhikalpan 2018 and 2019

OBSERVING EXPERIENCE

- PI of ALMA project “Zooming out: Disentangling the origins of large-scale structures around planet-forming disks” (project code: 2024.1.00524.S)
- PI of VLT project “Exploring effects of infall of material onto Class II systems” (project code: 113.26GY)
- Member of ALMA large programs DECO (project code: 2022.1.00875.L) and CHEER (project code: 2024.1.01001.L)
- Former member of “The JCMT Transient Survey”
- Assisted in optical observations at ARIES, India and Lulin Observatory, Taiwan

MENTORING EXPERIENCE

- Co-supervised Lauren Mason for the ESO Summer Research Programme (2024).
- Co-supervised Alessandro Ruzza for the ESO Summer Research Programme (2023).
- Supervised Jayden Burg, a high school student, for one week internship (Schülerpraktikum) (2024)
- Academic Helper for Physics and Mathematics at IIITDMJ Counseling Services (2016-2018)
- Student guide for the first year students at IIITDMJ (2016)

OUTREACH AND COMMUNITY SERVICE

- Refereed for Science journal
- Local Organizing Committee member for DECO All Team meeting at ESO, Germany (2024)
- Local Organizing Committee member for New Heights in Planet Formation conference at ESO, Germany (2024)
- Scientific Assistant for ESO Observing Programmes Committee (2024)
- Member of selection committee for ESO Summer Research Programme (2024)
- Co-organizer of Astronomy for Non-astronomers talks at ESO, Germany (2024-present)
- Co-organizer of Wine & Cheese seminars at ESO, Germany (2023-present)
- Chaired for Hypatia Colloquium 2022 at ESO, Germany (2023)
- Coordinator of [The Astronomy and Physics Society of IIITDM Jabalpur](#), India. We organized events like telescope nights, exhibitions, quizzes, etc. to promote astronomy among a variety of audiences. (2016-2019)
- E-Astronomer at [RAD@home](#) Astronomy Collaboratory, an Indian citizen-science research programme (2018-2019)